STACKS, DEFORMATION THEORY: LEMMA-FIRST-ORDER-DEFOS-COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 5802-5824 of sources/stacks-defos/pinned-original-source.tex; proof span is lines 5826-5921, the complete author proof (94 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 5 of queue_defos.json (pb=3834, stmt 5802-5824, proof 5826-5921).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0149: sources/stacks-defos/pinned-original-source.tex, source lines 5921–5921. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DIW: sources/stacks-defos/pinned-original-source.tex, source lines 5575–5588. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DIY: sources/stacks-defos/pinned-original-source.tex, source lines 5753–5768. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DIX: sources/stacks-defos/pinned-original-source.tex, source lines 5715–5723. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{O} \to \mathcal{O}_0$ be a surjection of sheaves of rings whose kernel is an ideal sheaf $\mathcal{I}$ of square zero. Let $K_0 \in D^-(\mathcal{O})$. A lift of K_0 is a pair (K, α_0) consisting of an object K in $D^-(\mathcal{O})$ and an isomorphism $\alpha_0 : K \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0 \to K_0$ in $D(\mathcal{O}_0)$.*

Date: stacks-project:defos.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(1) *Given a lift (K, α) the group of automorphism of the pair is canonically the cokernel of a map*
$$\mathrm{Ext}^{-1}_{\mathcal{O}_0}(K_0, K_0) \longrightarrow \mathrm{Hom}_{\mathcal{O}_0}(K_0, K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I})$$
(2) *If there is a lift, then the set of isomorphism classes of lifts is principal homogenenous under* $\mathrm{Ext}^1_{\mathcal{O}_0}(K_0, K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I})$.

Proof. An automorphism of (K, α) is a map $\varphi : K \to K$ in $D(\mathcal{O})$ with $\varphi \otimes_{\mathcal{O}} \mathrm{id}_{\mathcal{O}_0} = \mathrm{id}$. This is the same thing as saying that
$$K \xrightarrow{\varphi - \mathrm{id}} K \to K \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0$$
is zero. We conclude the group of automorphisms is the cokernel of a map
$$\mathrm{Hom}_{\mathcal{O}}(K, K_0[-1]) \longrightarrow \mathrm{Hom}_{\mathcal{O}}(K, K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I})$$
by the distinguished triangle
$$K \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{I} \to K \to K \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0 \to (K \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{I})[1]$$
in $D(\mathcal{O})$ and Derived Categories, Lemma 0149. To translate into the groups in the lemma use adjunction of the restriction functor $D(\mathcal{O}_0) \to D(\mathcal{O})$ and $- \otimes_{\mathcal{O}} \mathcal{O}_0 : D(\mathcal{O}) \to D(\mathcal{O}_0)$. This proves (1).

Proof of (2). Assume that $K_0 = K \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0$ in $D(\mathcal{O})$. By Lemma 0DIY the map sending a lift (K', α_0) to the obstruction $o(\alpha_0)$ to lifting α_0 defines a canonical injective map from the set of isomomorphism classes of pairs to $\mathrm{Ext}^1_{\mathcal{O}_0}(K_0, K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I})$. To finish the proof we show that it is surjective. Pick $\xi : K_0 \to (K_0 \otimes^{\mathbf{L}}_{\mathcal{O}_0} \mathcal{I})[1]$ in the Ext^1 of the lemma. Choose a bounded above complex $\mathcal{F}^\bullet$ of flat $\mathcal{O}$-modules representing K. The map ξ can be represented as $t \circ s^{-1}$ where $s : \mathcal{K}_0^\bullet \to \mathcal{F}_0^\bullet$ is a quasi-isomorphism and $t : \mathcal{K}_0^\bullet \to \mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I}[1]$ is a map of complexes. By Lemma 0DIX we can assume there exists a quasi-isomorphism $\mathcal{G}^\bullet \to \mathcal{F}^\bullet$ of complexes of $\mathcal{O}$-modules such that $\mathcal{G}_0^\bullet \to \mathcal{F}_0^\bullet$ factors through s up to homotopy. We may and do replace $\mathcal{G}^\bullet$ by a bounded above complex of flat $\mathcal{O}$-modules (by picking a qis from such to $\mathcal{G}^\bullet$ and replacing). Then we see that ξ is represented by a map of complexes $t : \mathcal{G}_0^\bullet \to \mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I}[1]$ and the quasi-isomorphism $\mathcal{G}_0^\bullet \to \mathcal{F}_0^\bullet$. Set
$$\mathcal{H}^n = \mathcal{F}^n \times_{\mathcal{F}_0^n} \mathcal{G}_0^n$$
with differentials
$$\mathcal{H}^n \to \mathcal{H}^{n+1}, \quad (f^n, g_0^n) \mapsto (d(f^n) + t(g_0^n), d(g_0^n))$$
This makes sense as $\mathcal{F}_0^{n+1} \otimes_{\mathcal{O}_0} \mathcal{I} = \mathcal{F}^{n+1} \otimes_{\mathcal{O}} \mathcal{I} = \mathcal{I}\mathcal{F}^{n+1} \subset \mathcal{F}^{n+1}$. We omit the computation that shows that $\mathcal{H}^\bullet$ is a complex of $\mathcal{O}$-modules. By construction there is a short exact sequence
$$0 \to \mathcal{F}_0^\bullet \otimes_{\mathcal{O}_0} \mathcal{I} \to \mathcal{H}^\bullet \to \mathcal{G}_0^\bullet \to 0$$
of complexes of $\mathcal{O}$-modules. Exactly as in the proof of Lemma 0DIW one shows that this sequence induces an isomorphism $\alpha_0 : \mathcal{H}^\bullet \otimes^{\mathbf{L}}_{\mathcal{O}} \mathcal{O}_0 \to \mathcal{G}_0^\bullet$ in $D(\mathcal{O}_0)$. In other words, we have produced a pair $(\mathcal{H}^\bullet, \alpha_0)$. We omit the verification that $o(\alpha_0) = \xi$; hint: $o(\alpha_0)$ can be computed explicitly in this case as we have maps $\mathcal{H}^n \to \mathcal{F}^n$ (not compatible with differentials) lifting the components of α_0. This finishes the proof. □